

Cash Got Your Tongue?

CEO Language-Uncertainty Around Undisclosed Cash Acquisitions

Sina Soleimanipour

MSc in Management (Finance), Thesis Defense

2026

Telfer School of Management, University of Ottawa

Supervisors: Dr. Ali Akyol, Dr. Harshit Rajaiya

Examiners: Dr. Shantanu Dutta, Dr. Rengong (Alex) Zhang

Motivation: a firm that knows, but may not say

A firm privately committed to an acquisition it has not announced sits in a **disclosure bind**:

- The pending deal can be **material** even while negotiations are preliminary (*Basic v. Levinson*, 1988).
- The firm **may stay silent**: no general duty to disclose merger talks.
- But once it speaks, it **may not mislead** (Rule 10b-5): a flat denial is not a lawful option.
- Theory: informed managers may rationally withhold (Verrecchia 1983), and silence can survive in equilibrium (Dye 1985).

The bind

The CEO must host the routine earnings call, cannot address the one material thing, and cannot deny it either. Does the spoken record track that withholding?

Research question: does residual CEO Q&A uncertainty track a deal's passage from *private* to *public*: elevated while a cash acquisition is withheld, receding once announced?

Preview of findings (all within-firm, correlational):

- Elevated in the quarter *before* a cash acquisition; no comparable rise detected before stock acquisitions.
- Indistinguishable from zero once the deal is announced, even before completion, while the funding cash stays until closing.
- The elevation concentrates in cash deals; the difference survives a formal pooled test.

A descriptive regularity, read through disclosure theory: no causal identification, no tested mechanism.

Measure: residual CEO Q&A uncertainty (UncResCEO)

Raw input: share of uncertain language in the CEO's unscripted Q&A answers on each earnings call.

Problem: executives differ persistently in how tentatively they speak. A level comparison mixes *style* with *state*.

Solution (decomposition following Dzielinski, Wagner, and Zeckhauser 2021): net out each executive's persistent speaking style; keep the **call-varying residual**.

- *UncResCEO* is centered near zero by construction; $SD = 0.3010$ (all-universe Table 1 basis).
- Computable only for CEOs with at least five calls at non-financial, non-utility firms (a coverage filter we disclose: it skews toward larger firms).

Details and estimating equations: backup slides.

Data and sample

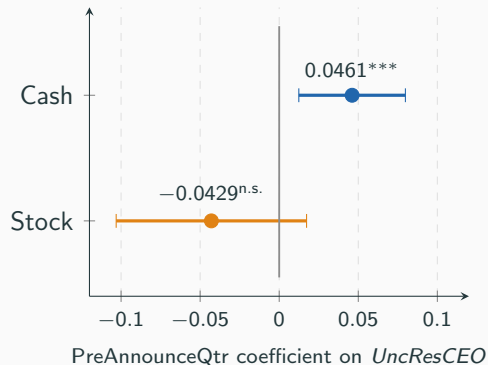
- **Earnings calls:** Capital IQ transcripts, U.S. public, non-financial, non-utility firms, 2002–2018.
- **Deals:** SDC acquisitions; cash arm = at least half cash, stock arm = at least half stock (managed comparison).
- **Language sample:** 88,205 calls across 1,884 firms.
- **Cash-ratio panel** is broader (2,232 firms): the cash ratio needs no speaking-style residual.
- $\text{CashRatio} = \text{cash and equivalents over assets}$; sample mean 0.1708.

Every table reports its own firm and observation counts; the uncertainty regressions describe a better-covered, larger-firm subset.

Empirical design: three main analyses

1. **MA1, run-up:** two-way fixed-effects regression of *UncResCEO* on a pre-announcement-quarter indicator; firm FE and calendar year-quarter FE. The firm is compared against itself.
2. **MA2, event time:** matched sample around the announcement; windows PRE2, PRE1, GAP (announced, pre-close), POST (completed); same outcome pair: uncertainty and cash ratio.
3. **MA3, cash-specificity:** both treatments pooled in one regression; the cash-minus-stock difference evaluated by a formal Wald test.
 - Standard errors clustered by firm; focal tests read **two-tailed**.
 - Post-announcement quarters dropped in the run-up design; never-acquirers form the baseline.

H1: pre-announcement run-up (cash vs. stock)



Cash acquirers: 0.0461^{***} (SE 0.0172; two-tailed $p = .0074$), about **15% of a residual SD**. Material but modest.

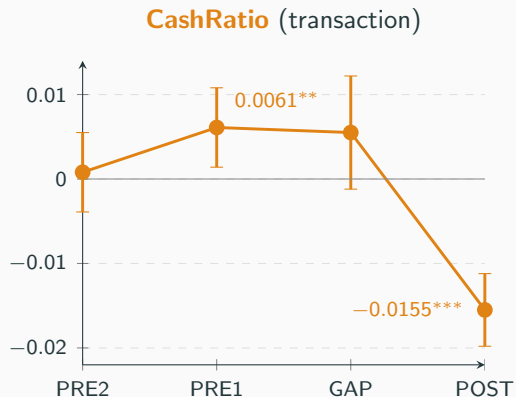
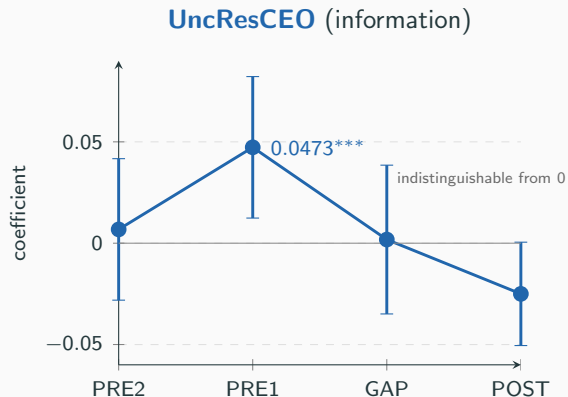
Stock acquirers: -0.0429 , not significant (SE 0.0307): a **noisy flat null**, not a detected decline.

The firm's own cash ratio also drifts up pre-announcement (0.0051^{***}).

Within-firm, correlational.

95% CIs; firm-clustered SEs. Table: run-up test, cols. (2), (6).

H1b: the round trip in event time (matched sample)



Cash acquirers, matched sample; $N = 28,102$ firm-quarters, 1,320 firms; 95% CIs, firm-clustered SEs. Two separate panels, one scale each.

H1b: two clocks

Information clock (UncResCEO)

- No pre-trend: PRE2 0.0068^{n.s.}
- Anticipatory rise: PRE1 0.0473^{***}
- Indistinguishable from zero once announced: GAP 0.0018^{n.s.}
- Drop PRE1–GAP = 0.0455^{**};
PRE1–POST = 0.0723^{***}

Transaction clock (CashRatio)

- Builds pre-announcement: PRE1 0.0061^{**}
- **Persists** while announced but unclosed (GAP)
- Leaves at completion: POST –0.0155^{***}; GAP–POST = 0.0210^{***}

Reading

Uncertainty resolves on the *information* clock (announcement); the funding cash resolves on the *transaction* clock (completion). The elevation is anticipatory, not a reaction to the announcement.

POST uncertainty sits slightly below baseline (–0.0250^{*}); we do not over-read that overshoot. Cash persistence at closing is partly mechanical.

H1a: concentration in cash deals (formal pooled test)

Pooled regression, both treatments, one
Wald restriction (two-tailed):

	UncResCEO
Pre-announce, Cash	0.0459** (0.0185)
Pre-announce, Stock	-0.0524 ^{n.s.} (0.0436)
Cash – Stock	0.0983** (0.0476), $p = .039$

What we claim: the run-up **concentrates**
in cash deals; the difference clears the
five-percent bar in a formal test.

What we do not claim:

- Not *strict* specificity: significance rides on an imprecise stock arm, so the test is **supportive rather than definitive**.
- The proposed *cause* is not cash-specific: the cash build-up difference is 0.0064^{n.s.} (matched). **The mechanism stays open.**

Ruling out one alternative; one channel check

Analyst cash-scrutiny (measured alternative driver). Analysts do ask more cash questions pre-announcement; is the run-up just that?

- Run-up survives adding CashScrutiny and its interaction: PreAnnounceQtr 0.0413***; interaction $-0.0056^{n.s.}$.
- Hedge kept as in the thesis: measured scrutiny **does not account for *this* run-up**; we do not claim scrutiny never matters (the interaction test is underpowered).

Bid-ask spread (information-asymmetry gauge). Two per-component facts:

- The residual is **unrelated** to the post-call spread (n.s. in all 12 specifications).
- The *scripted presentation* is positively associated with it (contemporaneous, one-tailed, within-firm).
- We report the two facts side by side; the between-component difference is **not tested directly**.

The pattern holds qualitatively under:

- ✓ **Withdrawal as resolution:** treating withdrawn deals' withdrawal date as the resolution event.
- ✓ **Dropping the cash equation's dynamic term** (lagged CashRatio partial adjustment).
- ✓ **All deals** (removing the first-deal restriction).
- ✓ **Logit** variants of the linear specifications.

Full robustness tables in the thesis appendix; key fragments in the backup slides.

Contributions (four, each descriptive)

To our knowledge, this thesis is the first to:

1. Read **residual Q&A uncertainty** in the anticipatory window before an acquisition is public (nearest prior work reads managed tone, strategy vocabulary, and prices).
2. Document that the pattern **concentrates in cash acquisitions**, a contrast that survives a formal pooled test.
3. Read the unscripted Q&A as tracking a deal's **passage from private to public**, connecting withholding theory with transaction-anticipation evidence, while claiming no causal channel.
4. Show the residual is **unrelated to the post-call bid-ask spread** while the scripted presentation is positively associated with it.

Limitations (by design)

- **Correlational, within-firm.** No causal identification; we characterize a regularity, we do not recommend acting on it.
- **Mechanism open.** Compliance-constrained speech and strategic reticence are observationally equivalent here; the cash build-up itself is not cash-specific (cause leg n.s.).
- **Stock arm is a noisy null**, not evidence of suppression; it is less precisely estimated (underpowered caveat).
- **Masking motivation** (stock acquirers manage the narrative) motivates the comparison; it is *not* identified.
- **Selection:** the residual needs five calls per CEO, so the language sample skews toward larger, better-covered firms.
- **Generated regressand:** the residual is a first-stage output; we address inference in Ch. 2 (backup slide).

Takeaways

1. A constrained executive's unscripted answers carry a readable, **anticipatory** trace of an undisclosed cash acquisition.
2. The trace is on the **information clock**: elevated before, indistinguishable from zero once announced; the funding cash stays until completion.
3. It **concentrates in cash deals** (formal pooled test); concentration, not strict specificity; mechanism open.

Thank you.

Questions welcome.

Backup slides

Backup: run-up test (MA1), focal columns

	Cash acquirers		Stock acquirers	
	CashRatio	UncResCEO	CashRatio	UncResCEO
PreAnnounceQtr	0.0051*** (0.0017)	0.0461*** (0.0172)	−0.0036 (0.0035)	−0.0429 (0.0307)
Firm FE, Year-Qtr FE	Yes	Yes	Yes	Yes
N (firm-quarters)	67,590	27,622	84,979	39,377
Firms	2,232	1,248	2,281	1,338
R^2	0.665	0.001	0.672	0.001

Firm-clustered SEs; controls included (leverage, size, Tobin's Q, ROA, capex, dividend dummy, scaled CFO; lagged CashRatio in the cash-ratio equations). Never-acquirers form the baseline; post-announcement quarters dropped. Two-tailed stars. Full table: thesis, run-up test.

Backup: event-time estimates (MA2), matched sample

	(1) UncResCEO	(2) CashRatio
PRE2 (pre-trend)	0.0068 (0.0178)	0.0008 (0.0024)
PRE1 (pre-announce)	0.0473*** (0.0178)	0.0061** (0.0024)
GAP (announced, pre-close)	0.0018 (0.0187)	0.0055 (0.0034)
POST (completed)	− 0.0250* (0.0130)	− 0.0155*** (0.0022)
Drop: PRE1 − GAP	0.0455**	0.0006
Drop: GAP − POST	0.0268	0.0210***
Drop: PRE1 − POST	0.0723***	0.0216***
N (firm-quarters); Firms	28,102; 1,320	28,102; 1,320

Firm and year-quarter FE; controls included; firm-clustered SEs; two-tailed stars.

Backup: cash-specificity test (MA3)

	EFFECT	PROPOSED CAUSE	
	UncResCEO (matched)	CashRatio (matched, +lag)	CashRatio (full panel, +lag)
Pre-announce, Cash	0.0459** (0.0185)	0.0056** (0.0025)	0.0049*** (0.0019)
Pre-announce, Stock	−0.0524 (0.0436)	−0.0008 (0.0066)	−0.0043 (0.0051)
Cash — Stock (Wald)	0.0983** (0.0476)	0.0064 (0.0071)	0.0092* (0.0055)
N; Firms	25,600; 1,191	24,347; 1,176	61,876; 2,141

Pooled: 837 first cash acquirers, 282 first stock acquirers; baseline = firms doing neither. Firm and year-quarter FE; firm-clustered SEs; two-tailed. EFFECT is cash-concentrated ($p = .039$); the proposed CAUSE (cash build-up) is not (matched leg n.s.), so the mechanism stays open.

Backup: constructing *UncResCEO*

1. For each call, measure the uncertainty share of the CEO's Q&A answers (uncertain-word counts over words, Loughran-McDonald style lexicon; Ch. 2, eq. for *UncAns*).
2. Decompose following Dzielinski, Wagner, and Zeckhauser (2021): regress the call-level measure on executive fixed effects (persistent speaking style) plus controls including the scripted-presentation uncertainty (*UncPreCEO*).
3. *UncResCEO* = the residual: the **call-varying** component of CEO answer uncertainty.

Why residual, not level: levels compare tentative speakers to confident ones; the residual compares each CEO to *their own* baseline.

Requires at least five calls per CEO. Convergent validity: the residual co-moves positively with political-risk and policy-uncertainty measures (one-tailed tests; economic magnitudes modest; disclosed in Ch. 2).

Backup: HighCash vs. HighCashScrutiny

Two distinct indicators; not interchangeable:

- $HighCash = \mathbf{1}[CashRatio \geq p67]$: top **tercile** of winsorized (1/99) CashRatio; sample mean 0.3333. Used as a regressor only in the scrutiny-validity table.
- $HighCashScrutiny = \mathbf{1}[CashScrutiny > \text{median}]$: the median is 0, so it flags **any** cash scrutiny ($\approx 11\%$ of calls; mean 0.1127).

Scrutiny measure is real: a firm's cash position predicts cash scrutiny (CashRatio 0.7530*** / 0.8519***; HighCash 0.1754*** / 0.1921***; one-tailed for the cash coefficient), yet scrutiny does not move the residual ($-0.0000^{n.s.}$ OLS; $-0.0003^{n.s.}$ logit, channel table).

$CashRatio$ = cash and equivalents (cheq) over total assets (atq).

Backup: generated-regressand inference

Concern: *UncResCEO* is a first-stage residual used as the dependent variable downstream (Pagan 1984 class of problems).

- Addressed in Ch. 2 (measure construction and estimating equations): the first stage is estimated on the full call panel, and downstream regressions treat the residual as data.
- Because the generated variable is the *regressand* (not a regressor), coefficient estimates downstream are not biased by the construction; the first-stage sampling error inflates the outcome's noise, which works *against* finding the run-up.
- Focal results are read two-tailed and survive (run-up $p = .0074$; timing and cash-specificity tables are two-tailed).

Backup: why a formal pooled test (not side-by-side stars)

Gelman and Stern (2006): the difference between “significant” and “not significant” is not itself statistically significant.

- So we **never** claim cash-specificity from “cash is starred, stock is not.”
- Instead MA3 pools both treatments and tests the **difference directly**: 0.0983 (SE 0.0476), $p = .039$, two-tailed Wald.
- That is also why the claim stays at *concentration*: the difference clears 5%, but its significance rides on the imprecise stock arm (-0.0524 , SE 0.0436), so: supportive rather than definitive.

Backup: why cash deals, not stock?

Motivation, not identification:

- A stock acquirer's currency is its own price, so it has an incentive to *manage the narrative* rather than fall silent; a cash acquirer has no such currency to defend, and secrecy protects the deal.
- This masking asymmetry motivates comparing cash against stock; the thesis **does not identify** it as the operative channel.

What the data say:

- Cash arm rises (0.0461^{***}); stock arm is a noisy flat null ($-0.0429^{n.s.}$): **the gap is cash rising, not stock suppressed**.
- Formal difference 0.0983^{**} ($p = .039$): concentration.
- The stock arm is less precisely estimated (SE 0.0307 vs. 0.0172): an underpowered-comparison caveat we keep.

Backup: cash-scrutiny detail

Three-step logic (all firm and year-quarter FE, firm-clustered):

1. **The confound is real:** scrutiny itself rises pre-announcement (HighCashScrutiny 0.0408**, run-up table col. 4).
2. **The measure is valid:** a firm's cash position predicts cash questions (CashRatio 0.7530*** / 0.8519***, one-tailed).
3. **But it does not carry the run-up:** adding CashScrutiny and its interaction leaves PreAnnounceQtr at 0.0413***; interaction $-0.0056^{n.s.}$ (CI roughly $[-0.027, +0.016]$).

Honest bound: a null-only rule-out; about 89% of calls draw no cash scrutiny, so the interaction is underpowered. We claim only that measured scrutiny does not account for *this* run-up.

Backup: robustness inventory

- **Withdrawal as resolution** (Ch. 4): withdrawn deals resolve uncertainty at withdrawal; pattern consistent.
- **Dropping the dynamic term:** cash-ratio equation without the lagged partial-adjustment term; conclusions unchanged.
- **All deals:** removing the first-deal restriction; pattern holds.
- **Logit variants:** binary high-uncertainty outcomes; consistent.
- Four robustness tables plus two logit tables in the thesis (`_robustness_tables.tex`).

Numbers quoted in the defense stay qualitative here; exact cells are in the thesis tables.